

The Silent Tax: Typo-Robustness Analysis (GSM8K)

DeepSeek-R1-Distill-Qwen-7B

All tables are *answered-only*: traces that hit the token cap before writing a final answer are excluded from the metrics. Configs are named `typo{rate}_real{ratio}`: *rate* = % of words corrupted, *real* = % of typos that form real words. Baseline is the clean input.

Accuracy and Flips

Table 1: Accuracy per config: answered-only and strict (truncated=wrong), with 95% bootstrap CI on acc_strict.

config	n	acc_answered	acc_strict	ci_lo	ci_hi
clean	20	0.7	0.7	0.5	0.9
clean_fix-spellcheck	1319	0.8407	0.8324	0.8112	0.8537
rate25_real0_fix-spellcheck	500	0.7867	0.76	0.724	0.798
rate25_real10_fix-spellcheck	500	0.7732	0.75	0.712	0.784
rate25_real20_fix-spellcheck	500	0.7651	0.736	0.696	0.774
rate25_real30_fix-spellcheck	500	0.7495	0.73	0.692	0.768
rate25_real40_fix-spellcheck	500	0.7377	0.72	0.68	0.756
rate25_real50_fix-spellcheck	500	0.7299	0.708	0.668	0.748
rate25_real60_fix-spellcheck	500	0.7479	0.724	0.682	0.762
rate25_real70_fix-spellcheck	500	0.7618	0.742	0.702	0.78
rate50_real0_fix-spellcheck	500	0.714	0.694	0.654	0.732
rate50_real10_fix-spellcheck	500	0.6934	0.674	0.634	0.714
rate50_real20_fix-spellcheck	500	0.6756	0.658	0.616	0.702
rate50_real30_fix-spellcheck	500	0.6687	0.65	0.608	0.69
rate50_real40_fix-spellcheck	500	0.6646	0.65	0.608	0.694
rate50_real50_fix-spellcheck	500	0.628	0.618	0.576	0.66
rate50_real60_fix-spellcheck	500	0.6196	0.606	0.564	0.648
rate50_real70_fix-spellcheck	500	0.6053	0.592	0.548	0.638
rate75_real0_fix-spellcheck	500	0.6562	0.63	0.588	0.674
rate75_real10_fix-spellcheck	500	0.6286	0.616	0.576	0.658
rate75_real20_fix-spellcheck	500	0.5825	0.572	0.53	0.616
rate75_real30_fix-spellcheck	500	0.5496	0.532	0.49	0.578
rate75_real40_fix-spellcheck	500	0.5456	0.538	0.494	0.582
rate75_real50_fix-spellcheck	500	0.4939	0.486	0.444	0.528
rate75_real60_fix-spellcheck	500	0.5155	0.5	0.458	0.546
rate75_real70_fix-spellcheck	500	0.4776	0.468	0.424	0.512
typo30_real0_fix-spellcheck	500	0.7778	0.756	0.716	0.792
typo30_real10_fix-spellcheck	500	0.7906	0.77	0.73	0.806
typo30_real20_fix-spellcheck	500	0.7791	0.762	0.724	0.798
typo30_real30_fix-spellcheck	500	0.7563	0.72	0.682	0.758
typo30_real40_fix-spellcheck	500	0.77	0.75	0.712	0.788
typo30_real50_fix-spellcheck	500	0.7438	0.72	0.68	0.756
typo30_real60_fix-spellcheck	500	0.734	0.712	0.672	0.752
typo30_real70_fix-spellcheck	500	0.7288	0.704	0.662	0.744
typo30_real10	20	0.55	0.55	0.35	0.75

Table 2: Answered-only flips vs clean, paired by question over n_both (answered in both conditions). r2w / w2r = questions correct on clean but wrong on this config / wrong on clean but correct on this config; r2w_ratio = r2w / n_both, w2r_ratio = w2r / n_both; net_flip_ratio = r2w_ratio - w2r_ratio = (r2w - w2r) / n_both, the net rate at which this config worsens answers relative to clean.

config	n_both	r2w	w2r	r2w_ratio	w2r_ratio	net_flip_ratio	mcnemar_b	mcnemar_c	mcnemar_stat	mcnemar_p
clean_fix-spellcheck	19	0	2	0.0	0.1053	-0.1053	0	2	0.5	0.47950012
rate25_real0_fix-spellcheck	19	4	2	0.2105	0.1053	0.1053	4	2	0.167	0.6830914
rate25_real10_fix-spellcheck	20	2	2	0.1	0.1	0.0	2	2	0.25	0.61707508
rate25_real20_fix-spellcheck	20	4	1	0.2	0.05	0.15	4	1	0.8	0.37109337
rate25_real30_fix-spellcheck	20	2	1	0.1	0.05	0.05	2	1	0.0	1.0
rate25_real40_fix-spellcheck	20	4	2	0.2	0.1	0.1	4	2	0.167	0.6830914
rate25_real50_fix-spellcheck	20	1	1	0.05	0.05	0.0	1	1	0.5	0.47950012
rate25_real60_fix-spellcheck	20	4	2	0.2	0.1	0.1	4	2	0.167	0.6830914
rate25_real70_fix-spellcheck	20	4	2	0.2	0.1	0.1	4	2	0.167	0.6830914
rate50_real0_fix-spellcheck	20	1	2	0.05	0.1	-0.05	1	2	0.0	1.0
rate50_real10_fix-spellcheck	19	3	2	0.1579	0.1053	0.0526	3	2	0.0	1.0
rate50_real20_fix-spellcheck	20	4	2	0.2	0.1	0.1	4	2	0.167	0.6830914
rate50_real30_fix-spellcheck	19	6	2	0.3158	0.1053	0.2105	6	2	1.125	0.28884437
rate50_real40_fix-spellcheck	20	7	1	0.35	0.05	0.3	7	1	3.125	0.07709987
rate50_real50_fix-spellcheck	20	6	1	0.3	0.05	0.25	6	1	2.286	0.13057002
rate50_real60_fix-spellcheck	20	6	1	0.3	0.05	0.25	6	1	2.286	0.13057002
rate50_real70_fix-spellcheck	20	6	1	0.3	0.05	0.25	6	1	2.286	0.13057002
rate75_real0_fix-spellcheck	20	5	3	0.25	0.15	0.1	5	3	0.125	0.72367361
rate75_real10_fix-spellcheck	20	7	0	0.35	0.0	0.35	7	0	5.143	0.0233422
rate75_real20_fix-spellcheck	19	2	2	0.1053	0.1053	0.0	2	2	0.25	0.61707508
rate75_real30_fix-spellcheck	20	5	1	0.25	0.05	0.2	5	1	1.5	0.22067136
rate75_real40_fix-spellcheck	20	9	2	0.45	0.1	0.35	9	2	3.273	0.07044043
rate75_real50_fix-spellcheck	20	7	2	0.35	0.1	0.25	7	2	1.778	0.18242244
rate75_real60_fix-spellcheck	19	7	1	0.3684	0.0526	0.3158	7	1	3.125	0.07709987
rate75_real70_fix-spellcheck	20	9	2	0.45	0.1	0.35	9	2	3.273	0.07044043
typo30_real0_fix-spellcheck	20	3	3	0.15	0.15	0.0	3	3	0.167	0.6830914
typo30_real10_fix-spellcheck	19	3	2	0.1579	0.1053	0.0526	3	2	0.0	1.0
typo30_real20_fix-spellcheck	20	2	2	0.1	0.1	0.0	2	2	0.25	0.61707508
typo30_real30_fix-spellcheck	18	2	2	0.1111	0.1111	0.0	2	2	0.25	0.61707508
typo30_real40_fix-spellcheck	19	0	3	0.0	0.1579	-0.1579	0	3	1.333	0.24821308
typo30_real50_fix-spellcheck	19	3	1	0.1579	0.0526	0.1053	3	1	0.25	0.61707508
typo30_real60_fix-spellcheck	19	3	1	0.1579	0.0526	0.1053	3	1	0.25	0.61707508
typo30_real70_fix-spellcheck	19	3	2	0.1579	0.1053	0.0526	3	2	0.0	1.0
typo30_real10	20	5	2	0.25	0.1	0.15	5	2	0.571	0.4496918

McNemar test (used in the flips table above).

- **What it is:** a paired significance test for before/after binary outcomes. It looks only at the questions that *changed* (the discordant pairs: b = right→wrong, c = wrong→right) and ignores those that stayed the same.
- **Null hypothesis:** typos break correct answers as often as they fix wrong ones ($b = c$, i.e. no net effect).
- **Statistic:** $(|b - c| - 1)^2 / (b + c)$ (column mcnemar_stat); mcnemar_p is its chi-square p -value with 1 degree of freedom.
- **When we reject it:** when mcnemar_p < 0.05 — the right→wrong vs wrong→right imbalance is too large to be chance.

Reasoning Length

Table 3: Absolute generated tokens per config (answered-only).

config	n_answered	mean_tok	median_tok	p90_tok	max_tok
clean	20	596.0	443.5	1467	2467
typo30_real10	20	603.2	517.0	1078	2140

Table 4: Mean generated tokens for correct vs wrong answers. $\text{wrong_over_correct} = \text{tok_wrong} / \text{tok_correct}$ (mean tokens of wrong answers divided by mean tokens of correct answers).

config	tok_correct	tok_wrong	wrong_over_correct
clean	629.9	516.7	0.8202
typo30_real10	664.5	528.2	0.7949

Self-doubt

The self-doubt markers are curated words grouped into two *banks*, each matched as a case-insensitive whole-word pattern in the reasoning text. Density is matches per 1000 reasoning words.

- **second_guess** — the proposal’s canonical self-correction words plus direct reversals: *wait, hmm, actually, reconsider, but wait, hold on, on second thought, let me recheck, double-check*.
- **uncertainty** — hedging / “not certain” words: *maybe, perhaps, possibly, might be, could be, not sure, I guess, presumably, or maybe / alternatively, unclear / confused*.

Example reasoning: “*Wait, maybe the total is 12... hmm, actually let me recheck*” contributes *wait, hmm, actually* (second_guess) and *maybe* (uncertainty). A separate *typo-noticing* bank (*typo, misspelled, doesn’t make sense, ...*) is treated as repair behaviour, not self-doubt.

Table 5: Doubt-marker density per config: second_guess and uncertainty, per 1k words and per trace.

config	n_answered	second_guess_per_1k	second_guess_per_trace	uncertainty_per_1k	uncertainty_per_trace	total_per_1k	total_per_trace
clean	20	5.773	1.15	3.514	0.7	9.287	1.85
typo30_real10	20	5.047	1.05	5.768	1.2	10.815	2.25

Table 6: Per-marker density (markers per 1000 reasoning words): clean vs pooled-typo. discrimination = typo minus clean (larger = more typo-responsive). All bank markers, ranked by discrimination.

marker	category	clean_per_1k	typo_per_1k	discrimination
maybe	uncertainty	1.757	2.403	0.646
actually	second_guess	0.0	0.481	0.481
unclear/confused	uncertainty	0.0	0.481	0.481
or maybe/alt	uncertainty	1.757	2.163	0.406
wait	second_guess	2.51	2.884	0.374
might be	uncertainty	0.0	0.24	0.24
could be	uncertainty	0.0	0.24	0.24
I assume	uncertainty	0.0	0.24	0.24
reconsider	second_guess	0.0	0.0	0.0
wait/actually no	second_guess	0.0	0.0	0.0
on second thought	second_guess	0.0	0.0	0.0
not sure	uncertainty	0.0	0.0	0.0
perhaps	uncertainty	0.0	0.0	0.0
possibly	uncertainty	0.0	0.0	0.0
I guess	uncertainty	0.0	0.0	0.0
presumably	uncertainty	0.0	0.0	0.0
not entirely	uncertainty	0.0	0.0	0.0
but wait	second_guess	0.502	0.24	-0.262
hold on	second_guess	0.502	0.24	-0.262
let me recheck	second_guess	0.502	0.24	-0.262
hmm	second_guess	1.255	0.961	-0.294
double-check	second_guess	0.502	0.0	-0.502

Table 7: Doubt density (markers per 1000 reasoning words) split by final outcome, per category (2g = second_guess, un = uncertainty, tot = both). For a group, density = total markers in that group / total reasoning words in that group x 1000. *_correct = over correct-answer traces; *_wrong = over wrong-answer traces; *_wrong_over_correct = _wrong / _correct.

config	2g_correct	2g_wrong	2g_wrong_over_correct	un_correct	un_wrong	un_wrong_over_correct	tot_correct	tot_wrong	tot_wrong_over_correct
clean	7.012	0.0	0.0	4.268	0.0	0.0	11.28	0.0	0.0
typo30_real10	6.947	0.0	0.0	7.608	0.879	0.115	14.555	0.879	0.06

Repair Behaviour

Table 8: How each corrupted word was handled, per config (categories defined below the table).

config	n_corrupt_words	silent_fix_n	flagged_n	misread_n	not_used_n	silent_fix_frac	flagged_frac	misread_frac	not_used_frac
typo30_real10	174	113	1	2	58	0.6494	0.0057	0.0115	0.3333

A *corrupted word* is a word that received a typo — it differs between the clean and typo question (e.g. *sum* → *sun*). We recover these by diffing the two questions, then place each into one bucket by which form appears in the reasoning:

- **silent_fix** — only the original word (*sum*) appears: a quiet recovery.
- **flagged** — both forms appear (*sum* and *sun*): noticed and fixed.
- **misread** — only the corrupted form (*sun*) appears: read as the wrong word.
- **not_used** — neither form appears in the reasoning.

Columns `*_n` are counts; `*_frac` are the share of `n_corrupt_words`.

Table 9: Word handling pooled by real-word ratio: the non-word vs real-word contrast.

real_ratio	n_corrupt_words	silent_fix_frac	flagged_frac	misread_frac	not_used_frac
10	174	0.6494	0.0057	0.0115	0.3333

Table 10: Misread rate split by final correctness. A typo’d word is ‘misread’ when the reasoning uses only its corrupted form (e.g. sun instead of sum). Within each outcome group, misread rate = (typo’d words that were misread) / (all typo’d words examined), pooled over the group. misread_correct = over problems answered correctly; misread_wrong = over problems answered wrong; wrong_over_correct = misread_wrong / misread_correct.

config	misread_correct	misread_wrong	wrong_over_correct
typo30_real10	0.0108	0.0123	1.148

Real-word Effect

This section isolates the effect of typo *kind* (real-word vs non-word), holding the question, corrupted positions, and rate fixed — something the vs-clean accuracy section cannot do. Three views:

- a **controlled paired** comparison of low-real vs high-real variants on the *same* questions, with a McNemar test (paired table);
- a **per-typo logistic regression** comparing the marginal harm of one real-word vs one non-word typo (logit table);
- whether wrong answers **flag the corruption** more or less as the real-word share rises (silent-failure table).

Read the signs, deltas and *p*-values in the tables below for this dataset’s result.

Table 11: Controlled paired comparison of two TYPO variants against EACH OTHER (baseline is the low-real variant, NOT clean – so this is not the vs-clean accuracy table). Within a fixed rate the same questions are corrupted in the same positions, only the typo KIND differs. Restricted to questions answered in both variants. acc_low_real / acc_high_real = accuracy on the lower / higher real-word variant; delta_acc = acc_high_real - acc_low_real; r2w = questions correct in the low-real variant but wrong in the high-real one; w2r = the reverse; mcnemar_p = McNemar test on that paired change. Negative delta with r2w $\hat{}$ w2r means real-word typos are more harmful than the non-word typos they replaced.

rate	compare	n_both	acc_low_real	acc_high_real	delta_acc	r2w	w2r	mcnemar_p
25	real10_to_real70	425	0.9176	0.9129	-0.0047	21	19	0.87436706
25	real10_to_real40	429	0.9138	0.9161	0.0023	17	18	1.0
25	real40_to_real70	433	0.9169	0.9007	-0.0162	30	23	0.40984673
50	real10_to_real70	399	0.8521	0.802	-0.0501	52	32	0.03816577
50	real10_to_real40	411	0.8564	0.8418	-0.0146	37	31	0.54428962
50	real40_to_real70	411	0.8345	0.8078	-0.0268	39	28	0.2218232
75	real10_to_real70	346	0.8237	0.7341	-0.0896	56	25	0.00085812
75	real10_to_real40	366	0.8033	0.7978	-0.0055	44	42	0.91412835
75	real40_to_real70	354	0.7994	0.7514	-0.048	42	25	0.05061718

Table 12: Logistic regression $\text{correct} \sim \text{num_real} + \text{num_nonword}$ over all answered typo traces ($\text{num_real} / \text{num_nonword} = \text{count of real-word} / \text{non-word typos in the problem}$). $\text{coef_}^* = \text{change in log-odds of a correct answer per one added typo of that kind (more negative = more harmful)}$; $\text{odds_}^* = \exp(\text{coef})$; intercept is the model constant. $n = \text{number of traces fit}$.

n	coef_num_real	coef_num_nonword	odds_real	odds_nonword	intercept
20	-0.55966	-0.0458	0.5714	0.9552	1.11723

Table 13: Among WRONG answers only, does the model flag the corruption, by real ratio? $\text{explicit_notice_frac} = \text{fraction of wrong-answer traces whose reasoning uses strong typo-flagging language (typo / misspelled / doesn't make sense / ...)}$; $\text{silent_frac} = 1 - \text{explicit_notice_frac}$. The proposal predicts real-word failures are quieter ($\text{explicit_notice_frac}$ should fall as real rises).

real_ratio	n_wrong	explicit_notice_frac	silent_frac
10	9	0.0	1.0

Lexical Grid (overview)

Table 14: All four marker families (second_guess , uncertainty , typo_noticing , repair_words) across configs.

config	second_guess	uncertainty	typo_noticing	repair_words
clean	5.773	3.514	0.753	0.502
typo30_real10	5.047	5.768	1.682	1.202

Spellcheck Recovery

Table 15: External spellcheck recovery by config. $\text{restored_exact_pct} = \text{percentage of typo words that were corrected back to exactly their original clean word}$.

config	n_questions	total_typo_words	restored_exact_n	changed_other_n	not_restored_n	restored_exact_frac	restored_exact_pct
rate25_real0_fix-spellcheck	500	6500	3500	1500	3000	0.538462	53.85
rate25_real10_fix-spellcheck	500	6500	4000	1000	2500	0.615385	61.54
rate25_real20_fix-spellcheck	500	6500	3500	1000	3000	0.538462	53.85
rate25_real30_fix-spellcheck	500	6500	3500	0	3000	0.538462	53.85
rate25_real40_fix-spellcheck	500	6500	3500	500	3000	0.538462	53.85
rate25_real50_fix-spellcheck	500	6500	3000	0	3500	0.461538	46.15
rate25_real60_fix-spellcheck	500	6500	2500	0	4000	0.384615	38.46
rate25_real70_fix-spellcheck	500	6500	2500	0	4000	0.384615	38.46
rate50_real0_fix-spellcheck	500	13000	9500	1000	3500	0.730769	73.08
rate50_real10_fix-spellcheck	500	13000	8500	500	4500	0.653846	65.38
rate50_real20_fix-spellcheck	500	13000	8000	1000	5000	0.615385	61.54
rate50_real30_fix-spellcheck	500	13000	6000	1500	7000	0.461538	46.15
rate50_real40_fix-spellcheck	500	13000	7000	500	6000	0.538462	53.85
rate50_real50_fix-spellcheck	500	13000	4500	1000	8500	0.346154	34.62
rate50_real60_fix-spellcheck	500	13000	5000	500	8000	0.384615	38.46
rate50_real70_fix-spellcheck	500	13000	4000	500	9000	0.307692	30.77
rate75_real0_fix-spellcheck	500	20000	12000	3500	8000	0.6	60.0
rate75_real10_fix-spellcheck	500	20000	11500	3500	8500	0.575	57.5
rate75_real20_fix-spellcheck	500	20000	12500	2000	7500	0.625	62.5
rate75_real30_fix-spellcheck	500	20000	8500	3500	11500	0.425	42.5
rate75_real40_fix-spellcheck	500	20000	9000	1500	11000	0.45	45.0
rate75_real50_fix-spellcheck	500	20000	8000	2000	12000	0.4	40.0
rate75_real60_fix-spellcheck	500	20000	6500	2000	13500	0.325	32.5
rate75_real70_fix-spellcheck	500	20000	5000	1000	15000	0.25	25.0
typo30_real0_fix-spellcheck	500	8000	4500	1500	3500	0.5625	56.25
typo30_real10_fix-spellcheck	500	7900	4940	1000	2960	0.625316	62.53
typo30_real20_fix-spellcheck	500	8000	5000	500	3000	0.625	62.5
typo30_real30_fix-spellcheck	500	8000	4500	500	3500	0.5625	56.25
typo30_real40_fix-spellcheck	500	8000	5000	0	3000	0.625	62.5
typo30_real50_fix-spellcheck	500	8000	5000	0	3000	0.625	62.5
typo30_real60_fix-spellcheck	500	8000	3000	500	5000	0.375	37.5
typo30_real70_fix-spellcheck	500	8000	3500	500	4500	0.4375	43.75

Table 16: Overall external spellcheck recovery for this dataset (pooled across spellcheck result files).

dataset	configs_count	total_typo_words	restored_exact_n	changed_other_n	not_restored_n	restored_exact_frac	restored_exact_pct
gsm8k	32	379900	186940	34000	192960	0.492077	49.21